Sum and Difference Formulas

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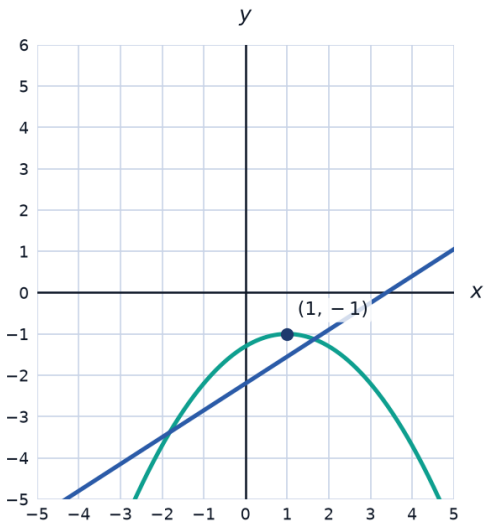
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Use a sum or difference formula to find the exact value of $\cos 75°$.

2 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Find the exact value of $\sin \frac{\pi}{12}$.



3 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$

b $\cos(A + B)$

c the quadrant of $A + B$

4 **[Foundational]** • **Target: Working with given values**

Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right)$

b $\cos(x - \pi)$

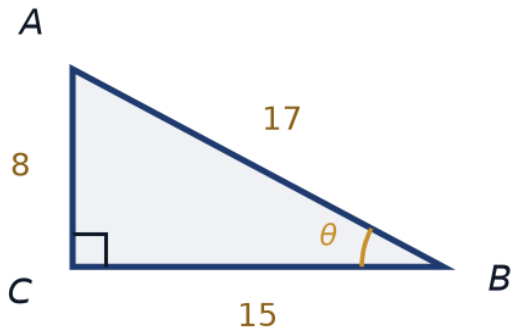
5 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.

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6 [Foundational] • Target: Verifying identities using sum and difference formulas

Show that $\tan\left(x+\frac{\pi}{4}\right)=\frac{1+\tan x}{1-\tan x}$ using the tangent sum formula $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$.



7 [Foundational] • Target: Applications

Two musical tones combine to form a sound wave modeled by $y=\sin\left(x+\frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Additional practice

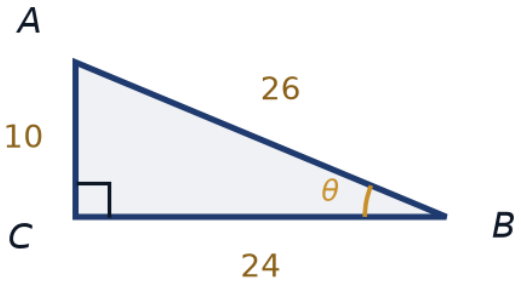
Angles A and B satisfy $\tan A=\frac{3}{4}$ (QI) and $\tan B=\frac{5}{12}$ (QI). Find $\tan(A+B)$ using the tangent sum formula.

9 [Intermediate] • Target: Additional practice

Simplify $\cos\left(\frac{\pi}{2}-x\right)\cos x+\sin\left(\frac{\pi}{2}-x\right)\sin x$ using the cosine difference formula in reverse.

10 [Intermediate] • Target: A worked derivation

Starting from the cosine difference formula $\cos(A-B)=\cos A\cos B+\sin A\sin B$, derive the cosine sum formula $\cos(A+B)$ by substituting $B\to -B$.



11 [Intermediate] • Target: A worked derivation

Find the exact value of $\sin 15°\cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta)=2\sin\theta\cos\theta$).

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12 [Intermediate] • Target: Sum and difference formulas with tables of values

Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A - B)$ exactly.

13 [Intermediate] • Target: Sum and difference formulas with tables of values

Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A - B)$ using the tangent difference formula.

14 [Intermediate] • Target: Expanding a general sinusoidal expression

Expand $\cos\left(x - \frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

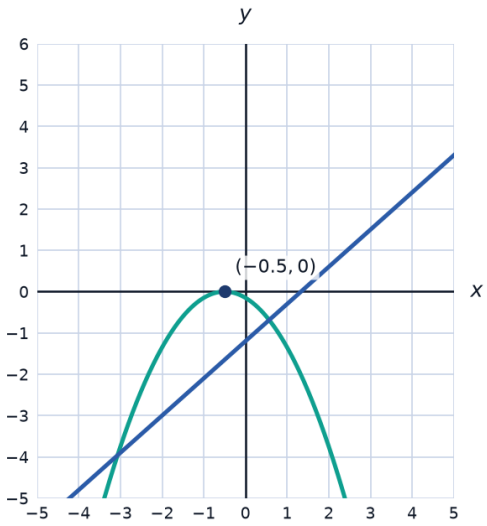
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 8.02 Graphing Rational Functions

Explain, using the unit circle, why $\cos(A - B)$ (not $\cos A - \cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

16 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 8.03 Introduction to Limits

Solve $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $x \in [0, 2\pi)$ by first expanding the left side.



17 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Given that a signal is modeled by $y = \cos\left(3x - \frac{\pi}{4}\right)$, expand it using the difference formula and identify the coefficients of $\cos 3x$ and $\sin 3x$.

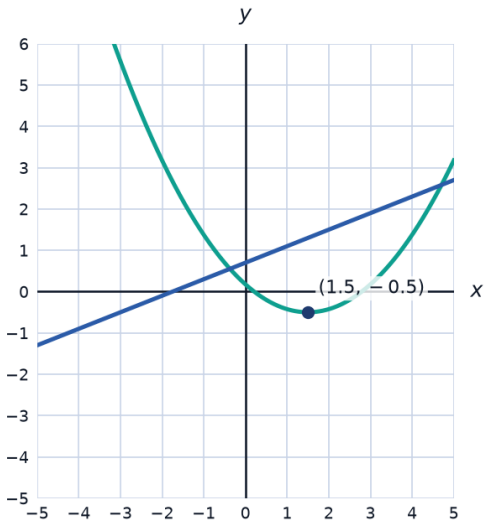
18 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 9.02 Geometric Sequences and Series

Two AC voltage signals are given by $V_1(t) = \sin(\omega t)$ and $V_2(t) = \sin\left(\omega t + \frac{2\pi}{3}\right)$ (phase-shifted by $120°$, common in three-phase power systems). Expand $V_2(t)$ fully using the sum formula, in terms of $\sin(\omega t)$ and $\cos(\omega t)$.

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19 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



20 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

A radioactive substance follows the decay model $M(t) = M_0e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exact values with sum and difference formulas

Find the 4th term of the geometric sequence with $a_1 = 2$ and $r = 3$.

a. 54

b. 162

c. 56

d. 57

22 [Foundational] • Target: Working with given values

Find the 3rd term of the geometric sequence with $a_1 = 4$ and $r = 3$.

a. 108

b. 40

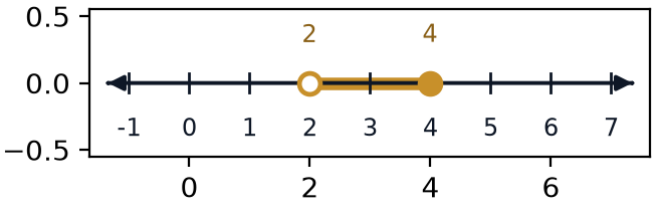
c. 36

d. 41

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23 [Foundational] • Target: Verifying identities using sum and difference formulas

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?



a. 4

b. 5

c. 9

d. 11

24 [Foundational] • Target: Applications

Find the term containing x^5y^1 in the expansion of $(2x + 3y)^6$.

a. $576x^5y^1$

b. $576x^1y^5$

c. $6x^5y^1$

d. $1152x^5y^1$

25 [Foundational] • Target: Additional practice

A hyperbola has vertices at $(\pm 3, 0)$ and foci at $(\pm 5, 0)$. Find b^2.

a. -16

b. 34

c. 16

d. 22

26 [Foundational] • Target: A worked derivation

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{5}{3}x$

b. $y = \pm\frac{3}{5}x$

c. $y = \pm 15x$

d. $y = \pm\frac{8}{5}x$

27 [Foundational] • Target: Sum and difference formulas with tables of values

What is the coefficient of the term containing x^3 in the expansion of $(1 + x)^4$?

a. 8

b. 6

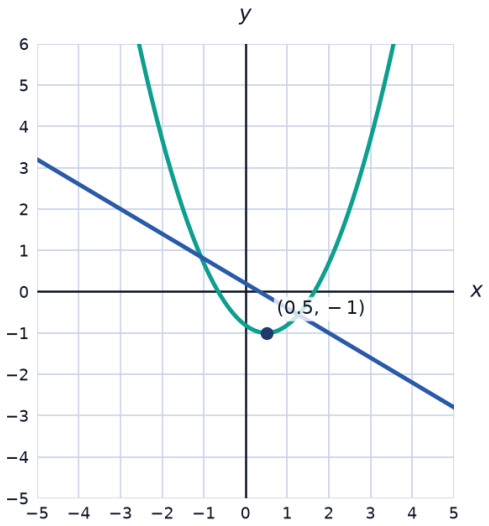
c. 1

d. 4

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28 [Foundational] • Target: Expanding a general sinusoidal expression

Find the foci of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.



a. $(\pm 5, 0)$

b. $(\pm\sqrt{34}, 0)$

c. $(\pm 2, 0)$

d. $(0, \pm\sqrt{34})$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A geometric interpretation

Find the end behavior of $f(x) = 3x^4 - x$ as $x \to +\infty$.

a. $f(x) \to 0$

b. $f(x)$ oscillates

c. $f(x) \to +\infty$

d. $f(x) \to -\infty$

30 [Intermediate] • Target: Solving equations using sum and difference formulas

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4

b. 6

c. 2

d. 5

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31 **[Intermediate]** • **Target: Exact values with sum and difference formulas**

Vector $\mathbf{u} = \langle 2, -3 \rangle$ is perpendicular to $\mathbf{v} = \langle 6, k \rangle$. Find k.

a. 6

b. -4

c. -3

d. 4

32 **[Intermediate]** • **Target: Working with given values**

Are $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle -3, 2 \rangle$ orthogonal, parallel, or neither?

a. Cannot be determined

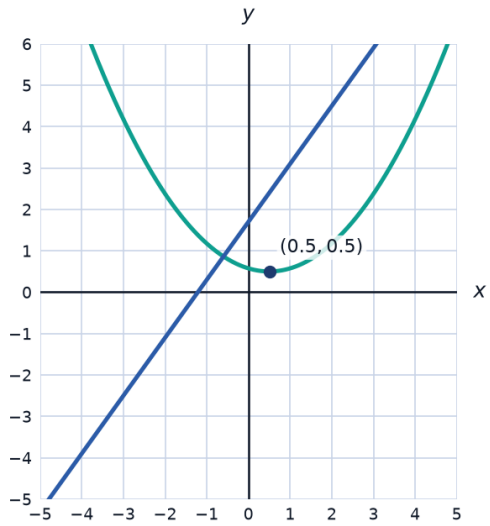
b. Orthogonal

c. Neither

d. Parallel

33 **[Intermediate]** • **Target: Verifying identities using sum and difference formulas**

Vector $\mathbf{v} = \langle 6, k \rangle$ is parallel to $\mathbf{u} = \langle 3, 4 \rangle$, with $\mathbf{v} = 2\mathbf{u}$. Find k.



a. -8

b. 6

c. 8

d. 10

34 **[Intermediate]** • **Target: Applications**

Find the eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. $\frac{9}{10}$

b. $\frac{3}{5}$

c. $\frac{4}{5}$

d. $\frac{5}{4}$

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35 [Intermediate] • Target: Additional practice

Find the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

a. 4

b. 5

c. 25

d. 6

36 [Intermediate] • Target: A worked derivation

For the parabola $y^2 = 8x$, what is the focus?

a. (4, 0)

b. (0, 2)

c. (-2, 0)

d. (2, 0)

37 [Intermediate] • Target: Sum and difference formulas with tables of values

Identify the conic section: $y = x^2 + 3$. (form 8)

a. Ellipse

b. Hyperbola

c. Circle

d. Parabola

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 9.02 Geometric Sequences and Series

For the limaçon $r = 1 + 1\cos\theta$, what is the maximum value of r?

a. 0

b. 3

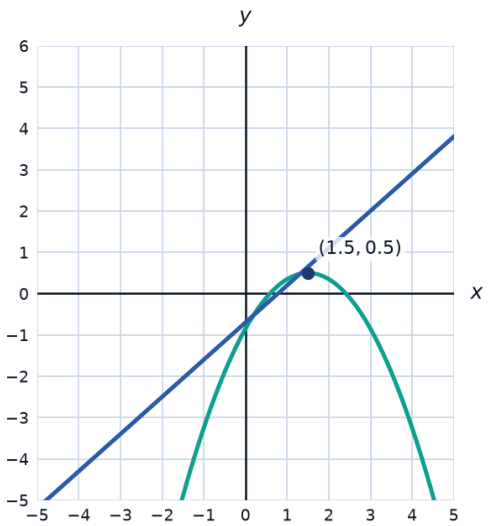
c. 1

d. 2

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39 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

A sequence is defined by $a_1 = 5$ and $a_n = a_{n-1} + 3$ for $n \geq 2$. Find a_6.



a. 20

b. 18

c. 17

d. 23

40 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

A vector $\mathbf{u} = \langle 3, k \rangle$ has magnitude 5. Find the positive value of k, given $k = 4$ is one solution.

a. 4

b. 2

c. 3

d. 5

41 [Challenging] • Target: Exact values with sum and difference formulas • Cross-Lesson: 1.02 Domain and Range

Write $\log_2(32) = 5$ in exponential form.

a. $2^5 = 32$

b. $32^5 = 2$

c. $2^{32} = 5$

d. $5^2 = 32$

42 [Challenging] • Target: Working with given values • Cross-Lesson: 1.03 Average Rate of Change

What is the vertex of the parabola $y = (x - 3)^2 + 4$?

a. (3, 4)

b. (3, -4)

c. (-3, -4)

d. (-3, 4)

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43 [Challenging] • Target: Verifying identities using sum and difference formulas • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the remainder when $f(x) = x^3 - 2x + 1$ is divided by $x - 2$.

a. 5

b. 7

c. 2

d. -5

44 [Challenging] • Target: Applications • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the degree and leading coefficient of $f(x) = 3x^4 - 2x^2 + 1$.

a. degree 4, leading coeff -3

b. degree 4, leading coeff 3

c. degree 3, leading coeff 3

d. degree 5, leading coeff 3

45 [Challenging] • Target: Additional practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Convert $\frac{5\pi}{6}$ to degrees.

a. 100°

b. 120°

c. 150°

d. 180°

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